

ROBOTICS: *Getting to know yourself as part of ECE*

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- 1. In your opinion, how would you define this area? What would be considered within this area and what is outside? This is a difficult question, and in some cases a widely accepted answer may not even exist.**

ANS: Robotics can be defined as an interdisciplinary field of creating and designing robots which, on their own or with slight human intervention, perform work or tasks. It uses ideas from mechanical engineering, electrical engineering, computer science, and artificial intelligence.

A robot is a machine especially one programmable by a computer capable of carrying out a complex series of actions automatically ^[1].

In my opinion robotics is a very vast field including machine building in such a way that it could perform tasks on its own or is easily operated by human beings. It includes skills in many field such as mechanical engineering, electrical engineering etc. even if we look at the SWISS-MILE robot, there were 6 types of engineers involved such as hardware engineer, AI engineer, mechanical engineer, electrical engineer.

- **Further sub fields included in robotics are:-**
 - Perception, learning, human and robot interaction.
 - Electrical Engineering.
 - Mechanical Engineering.
 - AI based Machine learning.
 - Machine and robot interaction.
- **Fields which might not be included in robotics are:-**
 - Data science and analytics.
 - Structural, Civil Engineering.
 - Other Purely software-based fields.

- 2. Give examples of work within this area that you find inspiring.**

ANS: Robotics inspires because it is an area that's about to make big leaps toward contributing to society and quality of life while every example shows the possibility for robotics to face challenges that come from the real world in the face of solving problems that would bring more safety, efficiency, health care, and accessibility to become significant change agents. Some projects which really inspired me are:-

- i. **Boston Dynamics Robots:** Boston Dynamics develops a series of dynamic highly mobile robots, including Big Dog, Spot, Atlas, and Handle. Since

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2019, Spot has been made commercially available, making it the first commercially available robot from Boston Dynamics ^[2].

- ii. **Surgical Systems:** Robotic surgery is a transformative advancement in healthcare, showing how robotics can bring unmatched precision, reduce human error, and improve patient outcomes.
- iii. **Self-Driving Cars:** It makes the cars self-driving in order to reduce possible traffic accidents and optimize routes for maximum fuel efficiency, then improves human accessibility to mobility, especially those who are not good at driving, so that makes this technology very much a difference in matters of public safety and impact on the environment.

The field of robotics has an incredibly wide array of projects and innovations that fall into many areas of engineering and science. Due to this broad scope, robotics has become more relevant because the world is in need of getting things done faster and in a more efficient way.

3. Why do you personally find this area exciting?

ANS: Since my childhood days, moving machines and robotics has always been a topic for me that would keep being fascinating. This is also the most interesting part to me: how machines are increasingly becoming highly complex and pretty sophisticated. It is awe-inspiring to see the sort of sophistication that a machine can generate and become independent with, the detail of activity it would possibly be able to perform, such things that no human or no human could ever even hope to accomplish on its own.

“The Guardian XO has been designed to target some specific (but also very common) types of jobs that require humans to repetitively lift heavy things. These jobs are generally not automatable, or at least not automatable in a way that’s cost effective—the skill of a human is required. These jobs are also labor intensive, which creates both short term and long term problems for human workers. Short term, acute injuries (like back injuries) lead to lost productivity. Long term, these injuries add up to serious medical problems for workers, many of whom can only function for between five and eight years before their bodies become permanently damaged.”^[3]

From this article and research we can confidently say that robotics is helping human beings at its peak, and what I really find exiting is that we humans now need to interact with robots which we designed. And it’s really very exciting to creating your own robot and designing it in such a way that it works for you, and interacts with you like a human being. Or at least works like that.

4. If you choose to delve deeper into this area, how would you prepare? You should begin by researching what particular courses in your curriculum would help you. What other knowledge and skills are needed? What resources are available online? Which faculty members at HU are interested in this area?

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ANS: For robotics it is essential for one to know the courses required for that. If I would delve deeper into robotics, I would first be looking at main courses required to stand out in all robotics engineer. Some common courses may include electronics, control systems, machine learning, mechanics and dynamics. Other than that I would be working on my other hard skills such as; learning programming languages and ROS (Robot Operating System).

At HU I would reach out to faculty members who have experience in robotics or are currently working on it. They could offer research opportunities and guidance to help me navigate complex topics. To be specific I will be looking for faculty members specialized in areas related to robotics such as; Electrical engineering, Computer Science, Mechanics, etc.

5. Recall that you had to fill an identity card at beginning of the semester that are still pasted on the lab window. Think back to that time and where you are now. Reflect and comment on the effect of this course (if any) on your first semester experience and future outlook.

ANS: When I came into Habib, there was never a moment where I doubted my course. It was something I selected on the basis of some good interest and not due to mere impulse, so I would enter lab work with not a single drop of hesitation, full of energy to learn as much as possible. This class has been an incredible experience so far, and even on my first day when I started, I already had that passion, so I hope to keep holding onto that enthusiasm till my 8th semester.

The course has helped clear up a number of questions I had prior and is sure to guide me further in the future. I have acquired the foundational skill set to build circuits and design structures. At first, when I entered Habib, I knew very little about the field, but the inspiration I gained was to continue learning, and hence I am ready to start research on subjects related to electrical, mechanical, and aeronautical engineering.

6. Think about your future career and list down some qualities that in your opinion will make your career a rewarding one.

ANS: It has always been my privilege to get hold of and be identified as one of the various bachelor's in electrical engineering of numerous Pakistani and other foreign degree holders. The facilitations and guidance from them are highly remarkable, as they highlighted their life experiences, so what is going to take me toward a very justified decision that would surely prosper in the long term.

As I mentioned earlier, I plan to focus specifically on enhancing my software and programming skills, especially as technology becomes increasingly digital and also on hardware like, building circuits etc. Additionally, I aim to graduate with a GPA of 3.5 or higher, which will open doors to top universities worldwide. For my personal development, I'll work on building my

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confidence by actively participating in various activities that will help improve my public speaking and presentation skills.

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